Please write your Enrollment Number) Enrollment No. 09801182024

(4)

End-Term Examination

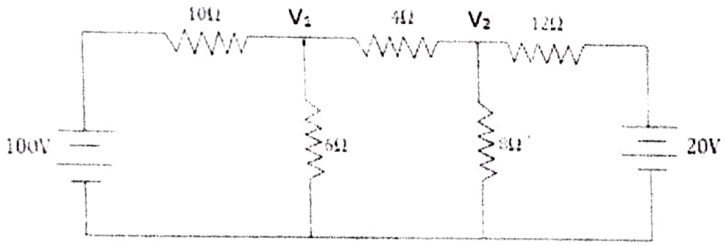
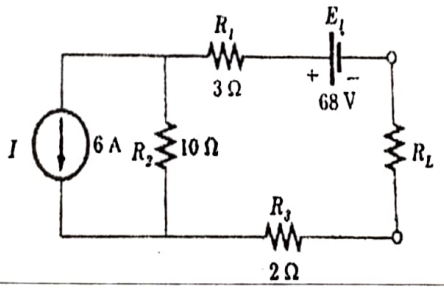
(CBCS)(SUBJECTIVE TYPE)(Offline)

Course Name: B.TECH : Sem: 2nd

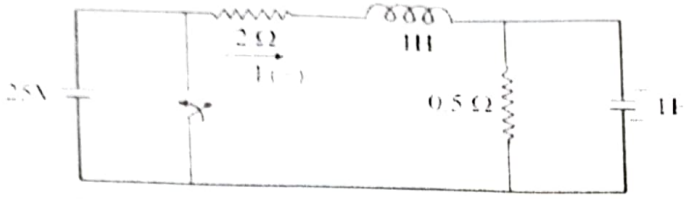
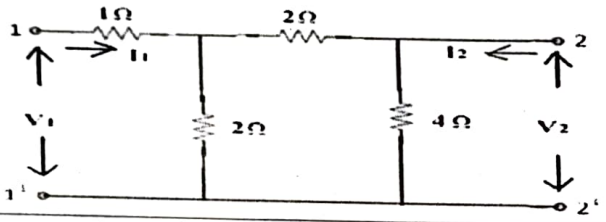
(May, 2025)

Subject Code: BEC 104	Subject: Network Analysis and Synthesis
Time :3 Hours	Maximum Marks :60

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1	Answer the following questions	(4*5=20)	CO Mapping
a	Two resistances when connected in series, the effective value of equivalent resistance is 100 Ohms. When connected in parallel the effective value is 24. Find the value of resistance R1 and R2.	5	CO1
b	Explain with the help of mathematical representation of the following theorems (i) Initial value Theorem (ii) Final value Theorem	5	CO2
c	Define y-parameters for a two port network and draw its equivalent circuit model.	5	CO3
d	How the function can be identified as a positive real function? Explain with the help of a suitable example	5	CO3
	UNIT I	**CO Mapping**	
Q2	Find Voltage V_1 and V_2 and current in 6 Ohm and 8 Ohm in the circuit shown below 	10	CO2
Q3	Find the value R_L in the figure for maximum power transfer to R_L and calculate the maximum power transferred. 	10	CO3

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	UNIT II		CO Mapping
Q4	Steady state is achieved in the circuit shown in figure with switch S as open. At t=0 the switch S is closed. Find the value of I(t) at t>0. 	10	CO2
Q5	A constant voltage source of 100V applied at t=0 to a series RC circuit with R=5 kΩ and C=20μF, given that there is no initial charge on the capacitor. Examine the current i and charge q for time t > 0.	10	CO2
	UNIT III		**CO Mapping**
Q6	Define h-parameters. Obtain the h-parameters of the circuit shown in the figure. 	10	CO3
Q7	(a) Drive the condition of symmetry and reciprocity for a two-port network in terms transmission (ABCD) parameters.	10	CO3
	UNIT IV		**CO Mapping**
Q8	What is an LC immittance function? State the properties of such functions.	10	CO4
Q9	Synthesize the network using Foster-I and Foster -II forms for the impedance function given below: $$Z(s) = \frac{s(s^2+9)}{(s^2+5)(s^2+13)}$$	10	CO4